

Introduction

I'm Shailesh. I bring 25+ years of experience building distributed systems, data platforms, and high-performance production infrastructure, with the last several years focused deeply on AI systems engineering — especially large-scale inference, multi-model serving, GPU-aware platforms, and low-latency distributed AI workloads.

My strongest area is building and optimizing production AI platforms where multiple models have to work together under strict latency, throughput, reliability, and cost constraints. I've worked on fraud-decisioning and identity platforms processing high transaction volumes, Siri-style conversational inference pipelines, and cyber-security inspection systems.

At the Principal level, my focus is designing AI infrastructure where performance, reliability, cost, and security are built into the same architecture .

Stories

- **CapitalOne Fraud Detection Platform**
 - **Innovation, Influence, Strategy , Execution and Communication**
 - 3-Tier Fraud Platform with ultra low latency Hot and Warm Path
 - **Conflicts**
 - Enabling GuardRail in warn path for LLM
 - **Failure**
 - Wrong Capacity estimation for LLM infra
 - **Difficult Issue**
 - Zero copy corruption
 - **Miscommunication, Cross team**
 - Arrow schema mismatch
 - **Mentorship**
 - Building team of RUST developers
- **Apple Siri HomePod**
 - **Innovation, Influence, Driving and Communication**
 - Unified Memory Architecture for multi-hop Siri Conversational pipeline
 - **Conflicts**
 - Ownership conflict for merging conversation stages
 - **Failure**
 - Using Milvus as choice for Vector DB . Added filter model to filter traffic
 - **Difficult Issue**
 - Fraud due to Rebuilding of IVF Indexes appending HSNW index
 - **Mis Communication Cross team**
 - Version mist match between Query and Document embedding models
 - **Mentorship**
 - Teaching HPC style of programming using C++

Delivery/Execution

- **Big Project with cross dependencies**
 - **Theme** :- Planning & Cadence
 - Program with Goals, KPI , Scope , Timelines , DRIs
 - Quarterly Roadmap with Milestone Plan for (2/4/6/ weeks) .
 - OKR (Business + Technical) with RAID (Risk, Assumptions, Issues and Dependencies)
 - Decision Records
 - **Story** :- 1 year owning Capital One's fraud-detection platform rebuild — migrating a legacy Java stack (35 ms p99) to a three-tier Hot / Warm / Cold path architecture (Rust/CUDA hot path < 5 ms, 8B-LLM warm-path reasoning, 70B-LLM cold-path triage) at 24,500 TPS, across 5 engineering teams(Infra, K8, Cloud, Search, Product), against a contractual card-network SLA.
 - **Action**
 - **North-star architecture with PRD + TDD** — wrote the product-requirements and technical-design docs; defined the hot/warm/cold tiers and the Arrow zero-copy contract between them _before_ any team built.
 - **OKR, roadmap & prioritization** — one KPI (sub-5 ms p99), a quarterly roadmap, milestone-level OKRs, each tier an independently shippable increment.
 - **Risk & dependencies; backup & contingency plan** — modeled cross-team dependencies as a DAG with a named DRI per edge and a 3-step escalation ladder; buffered the critical path 30% and kept an off-ramp at every milestone.
 - **Epics & stories** — broke each tier into epics and pull-able stories mapped to roadmap themes.
 - **Daily standup; bi-weekly / monthly status** — blockers-only daily sync across teams; bi-weekly written status for leads, monthly readout for directors.
 - **Executive communication** — one-page monthly execution readout (milestones, \$ impact, decisions needed) with explicit continue/adjust/stop calls.
 - **Result**
 - Each quarter shipped an **independently valuable milestone** — Hot, then Warm, then Cold path — each usable on its own.
 - Five teams stayed aligned on stable interface contracts; no milestone slipped the critical path.**
 - **Impact**
 - \$45M total savings in the first year.
 - Cut infra cost to ~1/3 → ~\$5M/year saved.
 - Fewer production incidents and faster triage** (p99 35 ms → 3.8 ms at 24,500 TPS)
 - What would have done **differently**

• **Ambiguity / Unclear Requirements / Difficult Target / Dependencies**

• Ambiguity / Unclear Requirements / Difficult Target / Dependencies

- **Theme** :- Turn a vague, hard-constrained brief into measurable scope — interview, spike the riskiest assumption
 - Program Board with dependencies (Owners, Due Date , Risk Levels)
 - Interface Contract , SLOs
 - DRI and Escalation Ladder
 - Weekly Status Check
 - Prioritization, Long Pole first , Low Hanging Items First
- **Story** : _Redesign Siri for HomePod in one quarter, under a 300 ms end-to-end latency budget, with no clear requirements on which use cases to support.
- **Action**
 - ****Interview stakeholders**** (Product + BAs) to enumerate probable use cases and define success metrics.
 - ****Decompose each use case**** into required inputs, service calls, and data dependencies.
 - ****Find the gaps and riskiest assumptions**, kick off a SPIKE immediately** (e.g., feasibility of ASR → NLU → retrieval → TTS inside 300 ms).
 - ****Prioritize use cases**** against the latency/timeline constraint and align with stakeholders.
 - ****TDD with phased delivery**** so use cases land incrementally.
 - ****Sign-off**** on scope and success metrics before building.
- **Result**
 - ****Delivered an MVP for the first 2 high-priority use cases**** (highest-volume, lowest-latency wins).
 - Demonstrated ****easier governance, faster iterations, and quicker triage**** than the prior process.
- **Impact**
 - Gave Product a ****firm launch date with a clear view of capabilities****.
 - Created a ****solid foundation for the next one-year plan****.
 - The architecture ****became a reference for other Apple product lines**** (Maps, Music).
- What would have done **differently**

• Scope Change

- **Theme** :- Scope Change , Push/Reject, Re-baseline
 - Impact Analysis
 - Change Request Process
 - Stake Holders Sign Off
 - Re-Baseline Dates and Risks
- **Story** :- Mid-program, Product wanted to route every transaction through the Warm Path (Tier-2, 13B reasoning LLM) — not just declines — to capture false negatives and generate training labels. At 24,500 TPS that means running an LLM on 100% of traffic: infeasible on latency and a budget explosion.
- **Action**
 - ****Quantified the impact**** in a change request: 13B reasoning on all transactions = $N \times$ the GPU fleet = \$X/month, and it blows the throughput/latency envelope.
 - ****Found the real need**** — they wanted false-negative _label coverage_, not literally every transaction reasoned.
 - ****Proposed an alternative****: smart sampling off the hot path — 100% of declines + the uncertainty band (scores near decision threshold) + a small random control sample — run asynchronously.
 - ****CR process**** — documented scope delta and cost of full vs. sampled, presented options A/B/C.
 - ****Stakeholder sign-off + rebaseline**** — agreed on sampling for v1; full coverage gated on v2 ROI review.
- **Result**
 - Captured ****~95% of the labeling value (FN recall) at ~5–10% of the GPU cost****.
 - Kept the warm path within SLA and the program within budget; no silent scope creep.
- **Impact**
 - ****Avoided a multi-fold GPU spend**** while still improving model recall.
 - Protected the launch timeline.
- What could have done differently

• Conflicts/Disagreement/Escalation and Stakeholder management

- **Theme** :- Present options with data, align on decision criteria, keep an escalation ladder ready — resolve at the working level. Trust
- **Story**:- Disagreement with the Cyber/Security team over using their centralized Guardrail Service for the warm path. Their mandate (PII/PAN filtering before the LLM) was non-negotiable for PCI-DSS, but the service added latency and a failure dependency that broke the warm-path budget.
- **Action**
 - ****Separated the requirement from the mechanism**** — compliance was the shared goal; _how_ was the conflict.
 - ****Measured the cost****: their service added ~X ms/call and a hard dependency threatening the 2–5 s budget at peak.
 - ****Presented options with data****: (A) Cyber's service as-is; (B) inline Sentinel PII filter certified against Cyber's ruleset; (C) hybrid — inline for hot/warm, central for async cold path.
 - ****Aligned on decision criteria**** — compliance non-negotiable; latency + availability weighted; had Cyber certify the inline filter by running their own test suite.
 - ****Escalation ladder on standby**** but resolved at the working level.
- **Result**
 - Inline filter ****passed Cyber's compliance suite**** and stayed within latency budget.
 - Conflict resolved ****without escalation**** — both teams owned the decision.
- **Impact**
 - Met ****PCI-DSS _and_ the latency SLA**** — both non-negotiables satisfied.
 - Cyber ****adopted the certified-inline pattern**** for other low-latency services.
 - Turned a blocking mandate into a ****reusable joint standard****.
- What could have done better

• Feature vs Tech debt management

- **Theme** :- Prioritization by business impact — quantify debt and features in the same currency, then sequence.
- **Story** :- Had to choose between paying down platform tech debt — GPU under-utilization (idle-cost waste) and observability gaps (slow triage) — versus porting a new Identity-Fraud product that carried time-to-market pressure.
- **Action**
 - ****Quantified the debt****: GPU utilization at ~X% → \$X/month wasted; observability gaps → 7 hour mean time to triage per incident

- **Quantified the debt**: GPU utilization at ~X% = \$1/month wasted; observability gaps = 2-hour mean-time-to-triage per incident.
- **Quantified the feature**: Identity-Fraud product's revenue/strategic value and deadline.
- **Mapped the interaction**: new product would ~2× platform load — onboarding onto an under-utilized, poorly observable base would amplify both debt costs.
- **Sequenced**: fix highest-interest debt the new product depends on first (Helios GPU scheduling + core observability), defer cosmetic debt, then port the product.
- **Made it one visible tradeoff** to leadership: "pay ~3 weeks of debt now to land the product on a stable base, or eat incidents + waste at 2× load."

• Result

- Landed **GPU scheduling + observability foundation**, then ported Identity Fraud onto a stable base.
- New product onboarded with the platform observable and well-utilized.

• Impact

- **GPU utilization ~70% → 82%+**, cutting idle cost; **faster triage**.
- Identity-Fraud product shipped with **fewer launch incidents**.

• What could have done better

• Critical Production bug/incident/Failure

- Theme
- Story
- Action
- Result
- Impact
- What could have done better

• Mentorship/Team Building/Caring for team

- Theme
- Story
- Action
- Result
- Impact
- What could have done better

People Management

• Team Topology

- Team Charter :- Mission, Customers , Success Matrix
- Org Map and Responsibility Matrix (Platform , Product , Enablement)
- Skill Matrix
 - **Q1**: Walk me through your **team charter**—mission, customers, success metrics—and how that drives **Platform/Product/Enablement** split.
 - *Answer hints*: Mission tied to business outcome (e.g., conversion↑, cost/tx↓); name customers; success KPIs (p95, cost/tx, incidents, TTI); Platform (serving/observability), Product pods (RAG/agents), Enablement (SDK/guardrails); API+SLO contracts; result = fewer stalls, faster onboarding.
 - **Q2**: How do you use a **skills matrix** to staff DRIs and close capability gaps?
 - *Answer hints*: Map streams → skills → DRIs; identify gaps (e.g., quantization); short-term contractor + pairing; checkpoint by sprint; measurable impact (e.g., INT4 safely, -30% cost, p95 < 200ms).

• Growth

- Individual Development Plan with Monthly/Quarterly outcomes .
- Delegation Ladder and Pairing (You + Leads + Seniors) . Vision/ Guardrails
- Mentorship and Sponsorship Roster .
 - **Q1**: How do you run **IDPs** with monthly/quarterly outcomes that lead to promotions?
 - *Answer hints*: IDP goals linked to KPIs; monthly demos, quarterly ship; evidence (ADRs, dashboards); influence & reliability signals; promotion dossier anchored in outcomes.
 - **Q2**: Describe your **delegation ladder** (you → leads → seniors) and use of **mentorship vs sponsorship**.
 - *Answer hints*: You set guardrails/vision; leads drive programs/RAID; seniors own components/DRIs; mentorship for skill, sponsorship for scope/visibility; scope expansions → title/comp readiness.

• Culture and Inclusion

- Meeting Hygiene :- Agenda , Rotating Facilitation
- Buddy System / Shadowing / Participation Equity
- Measure inclusion :- Speaking Time , Skip level pulses .
- Psychologic Safety :- Blame free , Critique Ides (Not People) , Condor with Care (Clear and Kind), Honest mistakes vs Bad Behaviors
- Recognition Cadence
 - **Q1**: What **inclusive meeting** practices do you enforce?
 - *Answer hints*: Pre-reads/agenda, rotating facilitation, time-boxed rounds, buddy/shadowing; decisions documented; shorter meetings, higher participation.
 - **Q2**: How do you **measure psych safety** and maintain a **recognition cadence**?
 - *Answer hints*: Speaking-time review, skip-level pulses; blameless PMs, "critique ideas, not people," clear & kind feedback; monthly spotlights tied to values; metrics: pages↓, pulse scores↑.

• Performance Management

- Performance Narrative Template :- Exceptions by Role and Level , Outcomes and Evidence linked , weekly checkpoints
- Improvement Plan :- Milestones , Enablement , Review Cadence , weekly checkpoints
- Feedback Log
 - **Q1**: Example of a **performance narrative** tied to role/level expectations.
 - *Answer hints*: Rubric-aligned competencies; evidence links (ADRs, KPIs, partner quotes); weekly checkpoints; decision

(meets/strong/exceeds) with next-scope plan.

- **Q2:** How do you run an **improvement plan** (PIP last resort) humanely?

- *Answer hints:* Expectations memo (outcomes/behaviors), milestones + enablement, weekly reviews; objective criteria; outcomes: improvement or fair transition with HR.

- Resilience

- Burnout Prevention , Individual Growth , Inclusion , Psychological Safety

- **Q1:** What systems prevent **burnout** while sustaining growth?

- *Answer hints:* On-call hygiene/runbooks, error budgets gate releases, meeting budgets, rotation of high-intensity streams; results: pages/person↓, cycle time↑.

- **Q2:** How do you spot **early burnout signals** and intervene?

- *Answer hints:* Signals: after-hours load, cycle time creep, missed 1:1s; actions: re-score with RICE, reduce WIP, shift staffing, protect focus time; recovery tracked over 1–2 sprints.

- Conflict Resolutions

- Conflict classification , Personality vs Tech

- Alignment n Shared Goal and Constraint

- Mediation notes with Options and Experiments

- Interest based negotiations and Nonviolent communication

- **Q1:** Resolve a **personality vs technical** conflict.

- *Answer hints:* Classify conflict; restate shared goals/constraints; time-boxed experiment; ADR with revisit date; coach behaviors; “disagree & commit.”

- **Q2:** Use **interest-based negotiation / NVC** to reach agreement.

- *Answer hints:* Surface needs behind positions; propose options addressing both (e.g., on-call simplicity + quality gains); phased rollout + observability; durable alignment.

- Communication and Leadership Cadence

- Communication on items from Delivery

- **Q1:** What is your **weekly communication system** (team → XFN → exec)?

- *Answer hints:* 5-bullet update (Goal, Progress vs Plan, KPI deltas, Risks, Asks); one shared dashboard; decisions captured as ADRs; avoids “green-until-red.”

- **Q2:** Example where cadence **prevented a surprise**.

- *Answer hints:* Early metric drift flagged; kill-switch/canary; fix, re-measure, resume; document in ADR; zero customer impact.

Product/Program Collaboration

- Problem Framing

- Problem Statement :- User Pain, Business Impact , Constraint etc .

- KPI Tree , Assumptions and Constraint

- Stakeholder map (PM, Tech, Infra, Financ , Support etc.)

- Discovery

- Current State Doc :- Architecture , Data Flows , Constraint etc.

- Research Brief :- User/Stakeholder interviews , log analysis , Support tickets

- Areas of improvement and Funnel analysis (5 Why)

- Opportunity sizing

- Proposal

- PRD or RFC

- Option comparison table (Option A/B/C with Cost, Latency, Risk, Time-to-ship).

- ADR (Architecture / Decision Record) with recommendation

- Draft rollout / launch plan (canary, shadow, regional)

- Budget / capacity estimate

- Stakeholder sign-off (PM, Eng, DS, Risk/Legal, Finance)

- Execution

- Items from Delivery

- Measurement

- Change Management

- Scope Change and Re-baseline from Delivery section

Project Management

- Management Challenges

- **Scope creep** → OKRs per pillar; Architecture Council gate.

- **Cost overruns** → FinOps dashboards, per-model budgets, autoscale/rightsizing.

- **Quality regressions** → pre-/post-deploy evals; feature flags; shadow tests.

- **Security incidents** → mandatory guardrails, DLP, audit logging by default.

- **Budget & FinOps**

- FP&A: **Workday Adaptive Planning, Anaplan, Pigment, Mosaic**. [WorkdayAnaplan IncPigmentMosaic](#)

- Cloud cost: **CloudZero** (unit economics, cost per feature/customer), **VMware Tanzu CloudHealth** (multi-cloud cost mgmt).

- [CloudZeroVMware](#)

- **Monthly Cost Reviews:** per-pillar unit cost and capacity targets.

- Resource Management

- **Quarterly Architecture Reviews** (AAR): team-pillar deep dives; approve standards.

- **SLO/SLA Baselines:** per service; green/yellow/red with auto rollbacks.

- **Security Gates:** pre-prod safety eval, PII redaction checks, model card updates.

- **Hiring / HRIS / ATS**

- HRIS: **Workday** (HCM), **BambooHR** (SMB). ATS: **Greenhouse, Lever**. Know what they track and how you partner with HR/recruiting. [Team](#)

o Performance & engagement

- o **Lattice, Culture Amp** for reviews, 1:1s, pulse/engagement, and calibration workflows. [LatticeCulture Amp](#)

o OKRs / strategy execution

- o **WorkBoard, Quantive (formerly Gtmhub), Microsoft Viva Goals**—how to cascade objectives and run a cadence. [WorkdayBambooHRRGreenhouse](#)

Engineering delivery & incidents

- **PagerDuty, FireHydrant**—on-call, runbooks, stakeholder comms, retros. [PagerDutyFireHydrant](#)
- Delivery analytics often tie back to **DORA** metrics. [Dora](#)

o Competition Roadmap

o Management Resources

- The Manager’s Path — day-to-day mechanics of tech leadership. O'Reilly Media
- An Elegant Puzzle — scaling systems, teams, and org mechanics. [press.stripe.com](#)
- Radical Candor — practical feedback culture. Radical Candor
- High Output Management — timeless ops & management fundamentals. Amazon
- Team Topologies — org design for flow. Team Topologies
- Measure What Matters — OKRs as an operating system (use judiciously). What Matters
- <https://www.manager-tools.com/manager-tools-basics>

o Frameworks & Tools

- **RICE** — Prioritization scoring: **Reach x Impact x Confidence ÷ Effort** to rank ideas/features.
- **RACI** — Responsibility matrix: **Responsible, Accountable, Consulted, Informed** roles for each task/decision.
- **ADRs** — **Architecture/Decision Records**: short, versioned docs capturing a decision, alternatives, and why/when.
- **CCB** — **Change Control Board**: forum that reviews/approves significant scope or design changes.
- **PERT** — **Program Evaluation & Review Technique**: estimates using **optimistic/likely/pessimistic** times to model schedules.
- **PACT** — **Consumer-driven contract testing** (e.g., Pact): verifies producer/consumer services honor their API contracts.
- **RAID** — Program log of **Risks, Assumptions, Issues, Dependencies** tracked with owners/dates.
- **SCQA** — Narrative framing: **Situation, Complication, Question, Answer** for clear, logical storytelling.
- **BLUF** — **Bottom Line Up Front**: lead with the conclusion/ask, then provide supporting detail.
- **JTBD** — **Jobs To Be Done**: product lens focusing on the user’s underlying “job” they’re hiring a solution to accomplish.

STAR

Story 1 – Capital One – Difficult Customer & GenAI Chatbot (Customer Obsession)

Category: Story 1 – Difficult Customer / Customer Obsession

Theme: RAG chatbot rollout for a very demanding Ops leader; GenAI, RAG, observability, data quality.

S – Situation

Capital One’s credit card support center was under pressure: long handle times, high transfer rates, and senior operations leadership was skeptical of “chatbot hype” due to a failed earlier bot. You were asked to lead a new **LLM + RAG agent** to deflect call volume for complex “fee, dispute, and rewards” questions. The Ops VP was openly hostile, citing prior bad CX and hallucinations.

- o Often customers bypassed the automated system by saying “agent” and directly sought live agent help.
- o Agents spent a lot of time manually searching through multiple tools (wikis, SharePoint, PDFs, policy docs) to answer even routine questions on products, fees, and policies.
- o The cost of the solution was high—both due to **agent handle time** and multiple **model / platform teams** maintaining separate stacks (NLP, rules engine, IVR, web chat).
- o New feature or policy rollouts required coordination across several teams (NLP, UI, backend services), making **time-to-market** for changes very slow.
- o There was **no end-to-end explainability or observability**—it was hard to see which intent fired, what knowledge source was used, and why the system failed for a given user.
- o Previous “chatbot” attempts had damaged stakeholder trust; Operations leaders were skeptical about another AI initiative unless it clearly improved CSAT and deflection.

T – Task

Own the end-to-end design and rollout of the GenAI chatbot:

- Prove **safe, grounded answers** using enterprise knowledge bases.
- Hit target of **15–20% self-service deflection** in 3 months.
- Win over the skeptical Ops VP and frontline managers by showing reliability, not just demos.

A – Action

- Designed a **hybrid RAG + multi-agent workflow** using LangGraph, Bedrock/OpenAI, and OpenSearch/PGVector, with:
 - o Tools for policy lookup, fee calculators, and account-agnostic flows.
 - o JSON-schema outputs (Pydantic) to keep responses structured and controllable.
- Worked closely with Ops to curate the **source-of-truth KB**:
 - o Flagged conflicting docs, outdated policies, and missing flows.
 - o Added metadata tags (product, region, channel) to improve retrieval precision.
- Implemented **AI observability**:
 - o OpenTelemetry traces, LangSmith runs, plus dashboards for: groundedness, escalation rate, latency, and hallucination flags.
 - o Weekly review with Ops on real transcripts and error cases.
- When the Ops VP wanted “full launch to 100% traffic” after a strong POC, you **pushed back**:
 - o Proposed a phased rollout (5% → 25% → 50%) with guardrails and auto-escalation thresholds.
 - o Backed with data: what failure modes might look like at scale and how we’d detect/mitigate.

R – Result

- Within 8–10 weeks:
 - o ~**22% deflection** on eligible intents,
 - o ~**35% reduction** in manual handle time on calls that started with the bot,
 - o “Unhelpful / misleading” responses cut to low single digits through RAG + guardrails.
- Ops VP became a champion; the earlier “failed bot story” was replaced by this success, and you were asked to replicate the pattern for other domains (disputes, loans).

Covers keywords:

- Difficult customer, Went above & beyond, Missed expectation / recovery, Prioritizing customers, Customer wanted X but needed Y, Pushing back on customer, Delivering bad news, Most impactful customer win

Story 2 – Apple – Ownership & Big Delivery Under Pressure (LLM Search & Reco)

Category: Story 2 – Ownership & Big Delivery Under Pressure

Theme: Siri search/recommendation pipeline with LLM augmentation under hard launch deadline.

S – Situation

At Apple, Siri’s knowledge graph search & recommendation stack was missing relevance targets ahead of a major OS release. A new LLM

At Apple, Siri's knowledge-graph search + recommendation stack was missing relevance targets ahead of a major OS release. A new **LLM-augmented query understanding and neural re-ranking** pipeline was prototyped, but the project was behind schedule, ownership was unclear, and multiple teams (search, KG, personalization) pointed fingers.

Situation :- Siri's search and recommendation experience was falling short of internal quality targets ahead of a major OS release.

- Existing pipelines relied heavily on **keyword / BM25 search and classic ML rankers**, which struggled with complex, conversational, or ambiguous queries.
- A new **LLM-augmented query understanding + neural re-ranking** prototype existed, but lived as a research POC with fragile scripts and no production readiness.
- Multiple teams were involved—Search, Knowledge Graph, Personalization, Infra—and ownership for “end-to-end delivery” was unclear, leading to delays and finger-pointing.
- Latency budgets for Siri were strict (multi-device, global footprint, voice UX), so naive LLM integration risked breaking **p95 latency SLOs**.
- Leadership had committed to **relevance improvements in this OS cycle**, so there was a firm deadline with high visibility, but the project timeline was already slipping.
- On-call teams lacked clear dashboards or metrics for the LLM-assisted pipeline, making it risky to deploy something that couldn't be properly monitored in production.

T – Task

Step up as the **de facto technical owner** and:

- Stabilize and productionize the LLM-assisted search pipeline.
- Hit **relevance and latency SLOs** in time for the OS launch.
- Coordinate 3–4 cross-functional teams in ~12 weeks.

A – Action

- Took **end-to-end ownership**:
 - Mapped all components: query logs → embeddings → KG search → LLM re-ranker → final answer.
 - Identified the critical path and removed nonessential “nice-to-haves” from v1.
- Re-architected the pipeline:
 - Introduced **hybrid retrieval** (BM25 + semantic + KG) feeding into a **cross-encoder re-ranker**.
 - Used a smaller, distilled LLM for re-ranking to **meet latency targets**.
- Built robust **data & evaluation loops**:
 - Offline: curated eval sets from query logs by intent, language, and device type.
 - Online: A/B tests with guardrails for latency and click-through degradation.
- Implemented **observability and SLO tracking**:
 - Exposed metrics (p50/p95 latency, recall@k, click-through) into unified dashboards.
 - Defined on-call runbooks and error budgets with the platform team.
- Negotiated scope with Product:
 - Pushed back on lower-priority personalization features that risked latency.
 - Kept v1 focused on “core answers right, fast, and safe”.

R – Result

- Delivered the pipeline **on time for launch**.
- Achieved:
 - ~**12–15% lift in top-1 answer precision**,
 - **p95 latency** within SLO despite LLM re-ranking,
 - Significantly fewer “no answer” / generic responses.
- Leadership recognized you as the **technical owner** for Siri's LLM-augmented search and reused the architecture for other verticals.

Covers keywords:

- Extreme ownership, Impossible deadline, Many priorities / juggling, Took over failing project, Outside responsibility, High-pressure delivery, Team falling behind, Most important project

Story 3 – Capital One – Ambiguity, Innovation & LLM Platform (Training + Inference)

Category: Story 3 – Ambiguity, Problem-Solving & Innovation

Theme: Designing a multi-tenant LLM training + inference platform in a very ambiguous environment.

S – Situation

Capital One wanted to move from ad-hoc LLM POCs on Bedrock/OpenAI to a **first-class internal LLM platform**: multi-tenant, GPU-efficient, and compliant. Requirements were vague: some teams wanted fine-tuning, others only retrieval-based inference; budget and GPU allocation were contested; no one had a clear reference architecture.

Situation :- Capital One was seeing a rapid increase in GenAI POCs across lines of business, but had no unified LLM platform.

- Different teams were separately calling external APIs (OpenAI/Bedrock) or spinning up their own GPU stacks, leading to duplicated effort, inconsistent guardrails, and rising costs.
- Requirements were **ambiguous and conflicting**: some groups wanted full fine-tuning and training, others only low-latency inference and RAG; security and compliance had strong opinions but no standard patterns yet.
- GPU resources (on-prem / cloud) were limited and expensive; there was **no shared mechanism** for scheduling jobs, tracking utilization, or enforcing cost governance across tenants.
- There was no “golden path” for **data → training → evaluation → deployment**; teams reinvented pipelines and MLOps patterns for each use case.
- Observability for training and inference was ad-hoc: different teams used different tools (logs, homegrown dashboards, basic CloudWatch metrics) with no unified view for leadership.
- Leadership wanted a clear **platform strategy** that balanced experimentation speed with compliance, performance, and predictable cost—but nobody had yet defined or owned that architecture.

T – Task

Create an end-to-end **GenAI platform architecture** and deliver the first working slice:

- Support both **training (fine-tuning / PEFT)** and **high-throughput inference**.
- Provide **observability, cost governance, and guardrails**.
- Reduce per-use-case reinvention of pipelines and infra.

A – Action

- Ran **discovery sessions** with product, platform, security, and data teams to clarify:
 - What models (LLAMA, Mistral, proprietary) and sizes they actually needed.
 - Which workloads truly required fine-tuning vs prompt/RAG solutions.
- Designed a **control-plane / data-plane** architecture:
 - Data-plane: GPU clusters with **Kubeflow Training Operator, KFServing/KServe, vLLM/TensorRT-LLM**, continuous batching, and quantization for inference.
 - Control-plane: central APIs for job submission, model registry integration, config-as-code, and chargeback.
- Implemented a **prototype pipeline**:
 - Used **LoRA/QLoRA** fine-tuning via PyTorch + DeepSpeed on curated data from Databricks (Bronze/Silver/Gold).
 - Deployed fine-tuned models to **vLLM** with dynamic batching, KV cache tuning, and OpenTelemetry metrics.
- Built **AI observability + cost dashboards**:

- Training: TFLOPs/GPU, \$/step, throughput, convergence metrics.
- Inference: tokens/sec, cost per 1k tokens, latency histograms, model utilization by team.
- Simplified complexity:
 - Codified “**golden paths**”: RAG-first for many use cases, fine-tuning only where necessary.
 - Published reference templates (YAML, notebooks, FastAPI wrappers) so teams could onboard quickly.

R – Result

- Platform reduced **time-to-first-POC** for new teams from months to weeks.
- Achieved **30–40% lower inference cost** via quantization + continuous batching without violating p95 latency.
- Leadership adopted your architecture as the **standard GenAI platform** roadmap.

Covers keywords:

- Worked with no clear direction, High ambiguity, Saw hidden problem, Designed new solution / innovation, Data-driven decision, Incomplete info decision, Challenged status quo, Simplified complex system

Story 4 – Apple – Leadership, Conflict & Influence (Privacy vs Data for LLM/RAG)

Category: Story 4 – Leadership, Conflict & Influence

Theme: Conflict between ML team and Privacy/Legal over what data can be used for LLM/RAG personalization & observability.

S – Situation

For Siri’s LLM-augmented personalization and RAG flows, your team wanted to leverage richer **query logs and device telemetry** to improve relevance, and more detailed logs for **LLM observability**. The Privacy and Legal teams pushed back hard on logging and data retention, worried about identifiable patterns and regulatory exposure. Product was stuck between “better personalization” and “risk”.

Situation :- Apple wanted to enhance Siri’s relevance and personalization by using LLMs and RAG over richer logs and device telemetry.

- ◦ The ML/search teams saw an opportunity to improve results by leveraging more detailed **query logs, context signals, and click/engagement data** for both training and online evaluation.
- ◦ Privacy and Legal teams, however, were extremely cautious about any central logging that could inadvertently capture sensitive or identifying patterns, especially for EU regions and child accounts.
- ◦ Observability for the new LLM/RAG flows required some level of **request tracing, error logging, and feature inspection**, which seemed at odds with strict privacy constraints.
- ◦ Product managers were caught in the middle: they wanted better Siri experiences and more robust metrics, but could not risk shipping anything that might violate Apple’s privacy commitments.
- ◦ Discussions often stalled: ML engineers felt blocked by “privacy red tape”, while Privacy/Legal felt ML was pushing for “unnecessary” data collection.
- ◦ Without a clear compromise, there was a real risk that **key LLM-augmented features would slip or ship with weak observability**, making it hard to maintain or improve them after launch.

T – Task

Influence cross-functional partners to find a **balanced design**:

- Maintain strong relevance and debug-ability for the LLM/search system.
- Respect strict privacy constraints (especially for EU/child accounts).
- Avoid delaying roadmap items due to gridlock.

A – Action

- Facilitated **multi-team workshops**:
 - Brought in ML, Privacy, Legal, Security, and Product to map **user journeys** and **data flows** end-to-end.
 - Separated “must-have signals for quality” from “nice-to-have but risky” logging.
- Proposed a **tiered data strategy**:
 - On-device aggregation + federated signals for some personalization features.
 - **Differentially private** summaries and heavily-aggregated metrics for observability.
 - Strict redaction / hashing for sensitive fields before logs reached central systems.
- Negotiated **guardrails and controls**:
 - Short retention windows and regional data residency for certain logs.
 - Config flags to disable advanced telemetry for specific geos or user segments.
 - Clear documentation for “what is collected and why”.
- Created **shared dashboards**:
 - Showed that even with constrained data, we could still measure relevance, latency, and error rates.
 - Built trust by demonstrating how we detect and correct quality regressions without user-level raw data.
- Coached junior engineers on **how to discuss these trade-offs** with non-technical stakeholders.

R – Result

- Reached an agreement that allowed your team to ship LLM-augmented features **without delays**, while satisfying Privacy/Legal.
- The pattern (tiered data, on-device aggregation, strict retention) became a **template** for future ML/LLM projects.
- You were seen as a **bridge** between engineering and compliance, not just a “model person”.

Covers keywords:

- Conflict with stakeholder, Influenced without authority, Disagreed with manager/stakeholder, Convincing team / getting buy-in, Coaching / mentoring, Giving hard feedback (on risk), Aligning misaligned teams

Story 5 – Symantec – Failure, Risk & Learning (Security ML Model & False Positives)

Category: Story 5 – Failure, Risk, Ethics & Learning

Theme: New deep-learning phishing model for web security causing harmful false positives; you own the mistake and fix.

S – Situation

At Symantec, you led development of a new **deep-learning URL/page-text model** (char-CNN/LSTM/transformer) to detect phishing and malicious sites from web telemetry. Early offline metrics looked fantastic (high recall, strong AUC), so the team pushed to roll it into an inline detection path. Shortly after limited rollout, several large enterprise customers complained about **legitimate login portals being blocked**, disrupting their business.

Situation :- At Symantec/BlueCoat, you led development of a new deep-learning model to detect phishing and malicious sites from large-scale web proxy telemetry.

- ◦ The existing rules + reputation-based system had **good precision but missed newer, fast-changing phishing patterns**, so leadership wanted a modern deep learning model (char-CNN/LSTM/transformer).
- ◦ Offline experiments on historical data showed impressive metrics (high recall, strong AUC), creating optimism that this model could significantly boost protection.
- ◦ Under pressure to differentiate the product and close competitive gaps, there was a push to move the model quickly into an **inline blocking path**, not just analysis mode.
- ◦ Evaluation focused heavily on aggregate metrics; customer-specific validation sets and real-world “good” login patterns were underrepresented in the training/validation pipeline.
- ◦ The team rolled the model out in a limited but **real traffic path**, with partial safeguards—but without a fully mature shadow deployment and rollback framework.
- ◦ Shortly after rollout, large enterprise customers reported that legitimate login portals and internal web apps were being mistakenly blocked, causing business disruption and escalations to executive level.

T – Task

Own the failure, minimize customer impact, and **rebuild trust** while fixing the model and process:

- Diagnose why the model behaved differently in production.
- Reduce false positives to acceptable levels.
- Put processes in place so this kind of incident doesn’t repeat.

A – Action

- Immediately recommended **rolling back** the model from blocking mode to **shadow mode**:
 - Communicated transparently with Product and Support: what happened, what we know, what we’re doing.
- Led a **post-mortem** with detailed analysis:
 - Discovered training data bias (over-representation of certain “login-style” templates from previous campaigns).
 - Realized that some URLs mis-labeled in historical data caused the model to over-weight particular lexical patterns.
- Built a more robust **evaluation framework**:
 - Constructed customer-specific validation sets from their domains and known good/bad traffic.
 - Added business-critical whitelists and **tiered decisioning**: model → reputation DB → policy rules.
- Retrained and re-integrated the model:
 - Cleaned labels, added harder negative examples, and introduced **calibrated thresholds per segment**.
 - Deployed it first in **monitor-only mode** with rich telemetry logs to compare old vs new decisions.
- Documented **lessons learned**:
 - Mandatory customer-representative datasets before inline deployment.
 - Requirement: run models in shadow mode for N weeks and pass drift/fp criteria before gating real traffic.
 - Shared the learning with other ML teams to prevent similar issues.

R – Result

- False positives for those customers dropped to **below previous baselines** after retraining and gating.
- Customers appreciated transparent communication and the speed of rollback + fix; no major churn.
- The new deployment and evaluation policies became **standard practice** across multiple security products.
- You had a strong, honest “failure story” showing ownership, ethics, and learning.

Covers keywords:

- Major failure, Biggest mistake, Risk that went wrong, Missed goal / deadline (quality goal), Critical feedback received, Not proud of performance, Tough ethical decision (rollback, owning impact), Admitting you were wrong

Project :- Conversational service backend API for C1 CardTech team to help Customer and Agents

- **Problem :-** Capital one has the Legacy Conversation API build with legacy NLP solutions which was not working very well and have an underlying issues
 - Often customer bypass the automated system and seek and live agent help
 - Agent used to take lot of time to find the relevant information on Products and Policies to answer customer queries ,
 - Cost of the solution was too high accounted for both Agent time and multiple model teams .
 - New feature rollout was very time consuming due to multi team dependencies
 - No end to end Explainability .
- **Business Objective :-** Build the chatbot for C1 customer support with following objectives :-
 - Reduce the tripping from automated system to Agents by 50 % in Phase-1
 - Reduce the response time by providing enhanced Assistant to agents from 1-3 min to 10-30 sec
 - Reduce operation cost by reorganizing the teams and services .

Q & A

- Where do you see technology going in next 3/5 years
- What is the next big thing after GenAI

Leadership Principal	Definition and actions	Question	Situation	Task	Action Result
Customer Obsession		Who was your most difficult customer?			
		Give me an example of a time when you did not meet a client’s expectation. What happened, and how did you attempt to rectify the situation?	GDPR complainace for romaing users from EU		
		When you’re working with a large number of customers, it’s tricky to deliver excellent service to them all. How do you go about prioritizing your customers’ needs?			
		Tell the story of the last time you had to apologize to someone (customer).	GDPR complainace for romaing users from EU		
		Tell me about a time a customer wanted one thing but you felt they needed something else. What was the situation and what was the action you took?	GDPR complainace for romaing users from EU		
Ownership		Tell me about a time when you had to leave a task unfinished.	Story of Openstack implementation, where we wanted to build appliance cloud but couldn’t do it because virtual images was only supporting ZEN, and openstack	It was quite challenging , as several people has to work on several different area , which could meet the deadline .	But it triggered the new idea of creating common operating environment for all appliances .

			and openstack was on KVM.		
		Tell me about a time when you had to work on a project with unclear responsibilities.			
		Provide an example of when you personally demonstrated ownership.	Story of wrong DLP URL by customer	Collected list of such customer , send them an email with the help of Technical Writer .	Customer were very happy that we have proactively catch that , and help them to configure it correctly . Later on I came to know that this incident help Sales team to renew customer license very easily without much perusing .
		Tell me about a time when you took on something significant outside your area of responsibility. Why was it important? What was the outcome?	Story of wrong DLP URL by customer	Collected list of such customer , send them an email with the help of Technical Writer .	
		Describe a project or idea (not necessarily your own) that was implemented primarily because of your efforts. What was your role? What was the outcome?	Story of wrong DLP URL by customer . Story of creating COE , where each appliance has its own OS	Eliminate need of configuring DLP URLs , and check in our database instead of configuring it in Portal . Standardize OS , and most basic things will be part of Framework .	No more mismatch . Now more emphasis is on COE, and org approved lot of funding to hire new people for team .
invent and simplify		Tell me about a time when you invented something.	Story of SALT proxy minion, where conventional CM tools didn't work .	Started new module SALT proxy minion partnership with SaltStack	That project helps us as well as SALT team to sell their product to other customer , and they have granted free Enterprise License to us for lifetime .
		What improvements have you made at your current company?	Story of SALT proxy minion, where conventional CM tools didn't work .	Started new module SALT proxy minion partnership with SaltStack	That project helps us as well as SALT team to sell their product to other customer , and they have granted free Enterprise License to us for lifetime .
		Tell me about a time when you gave a simple solution to a complex problem.	Story of SALT proxy minion, where conventional CM tools didn't work .	Started new module SALT proxy minion partnership with SaltStack	That project helps us as well as SALT team to sell their product to other customer , and they have granted free Enterprise License to us for lifetime .
		Tell me about a time you had to think outside the box (think creatively) to close a sale or sell your product.	Story of SALT proxy minion, where conventional CM tools didn't work .	Started new module SALT proxy minion partnership with SaltStack	That project helps us as well as SALT team to sell their product to other customer , and they have granted free Enterprise License to us for lifetime .
		What is the most innovative project you've worked on?	Story of SALT proxy minion, where conventional CM tools didn't work .	Started new module SALT proxy minion partnership with SaltStack	That project helps us as well as SALT team to sell their product to other customer , and they have granted free Enterprise License to us for lifetime .
Are Right, A Lot	Leaders are right a lot. They have strong judgment and good instincts. They seek diverse perspectives and work to disconfirm their beliefs.	Tell me the time you made a mistake	Over Engineering with EU GDPR compliance , where we created EU cluster everywhere without properly		

			understanding GDPR . Didn't consider roaming customer and build dedicated cluster in each region .		
		Tell me the time you have disagreement with you colleagues or Boss	Story of choosing AWS Lambda		
		Tell me the situation when you have to make the decision in the absence of data .			
Deliver Results	Leaders focus on the key inputs for their business and deliver them with the right quality and in a timely fashion. Despite setbacks, they rise to the occasion and never settle	Tell me the challenging situation while working on a project and what you did to overcome it.	Story of SALT proxy minion, where conventional CM tools didn't work .		
		How you check your progress against your goals?			
		Tell me about a time where you not only met the goal but considerably exceeded expectations. How were you able to do it?	Story of SALT proxy minion, where conventional CM tools didn't work . Also, customize it in such away and made it generic to use for any other customer .		
		What's the most complex problem you've ever worked on?	Story of SALT proxy minion, where conventional CM tools didn't work .		
		Have you ever worked on something really hard and then failed?	DLP Lambda, lead to throttling problem and failure . Lesson learned to have contingency plans .		
Earn Trust	Leaders "Earn Trust" when they "take the hit." When undesirable outcomes happen, we're all quick to point the finger. If your team members see that you're willing to take the blame for the good of the team, even if it's not directly your fault, then they'll start to let go and trust you. As leader of a team, you need to accept the responsibility for both the good and the bad.	Building trust can be difficult to achieve at times. Tell me about how you've effectively built trusting working relationships with others on your team.			
		Describe a time when you significantly contributed to improving morale and productivity on your team.			
		Tell me about a piece of direct feedback you recently gave to a colleague. How did he or she respond?			
		Tell me about a time when someone (peer, teammate, supervisor) criticized you about a piece of work/analysis that you delivered. How did you react? What was the outcome?			
		Tell me about a time when you had to tell someone a harsh truth.			
		How do you convince someone who is resistant to what you're trying to do?			
Insist on Highest	Having high standards	Tell me about a time when	Following two		

Standards	means you make exceptionally high demands of yourself and the products and services you work on. At Amazon, standards are set through service level agreements (SLAs). An SLA is a set of agreed upon standards at which any service or product will perform. In an Amazon SLA, even the worst outcome will outpace industry standards.	you've been unsatisfied with the status quo. What did you do to change it? Were you successful?	points.		
	- Set SLAs for everything, and don't take shortcuts on instrumentation. - Continually self-critique your work to make sure the quality is the best it can be.	Tell me about a time you wouldn't compromise on achieving a great outcome when others felt something was already good enough. What was the situation?	Story of customer not using DLP efficiently, you place the system in place to detect such customers and proactively helped them to configure the URL .		
	- Accept and seek coaching and feedback from your manager and others about improving the quality of your work. - Demand that your team delivers high-quality products, services, and solutions.	What measures have you personally put in place to ensure performance improvement targets and standards are achieved?	Not only we catch the above customers , but also placed the system in place that without them being configured anything will work for them .		
	Coach employees about setting their own high standards and exceeding customer expectations.	Describe the most significant, continuous improvement project that you've led. What was the catalyst for this change and how did you go about it?	Above two points		
		Give me an example of a goal you've had where you wish you had done better. What was the goal and how could you have improved on it?	Story of diff salt minion port , which could have been salt with Proxy .		
Think Big	Leaders create and communicate a bold direction that inspires results. They think differently and look around corners for ways to serve customers.	Tell me about a time you took a calculated risk in order to achieve a professional goal. What were the tradeoffs? What was the outcome?	Salt Proxy Minion Story		
	Taking a radical approach and risks when necessary, always questioning traditional assumptions in pursuit of the best idea.	Tell me about a time you took a big risk and it failed. What did you learn? What would you do differently?	DLP Lambda, lead to throttling problem and failure . Lesson learned to have contingency plans .		